

Deep Reinforcement Learning

Optimal Control & Feedback Control

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Content

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- Recap of model-free RL
- Open-loop optimal control
 - Trajectory planning
 - Solution approaches
 - Stochastic optimal control
 - The linear case
- Closed-loop control
 - Model Predictive Control (MPC)
 - Differentiable Predictive Control (DPC)

Where are we?

Chapter	Topic	Content
	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6-13	DQN, policy gradients, actor-critic, PPO/SAC, exploration	Deep RL from basics to modern algorithms
	Model-Based Control	
14	Optimal control & feedback control	How to do control when we know the model
	Model-based reinforcement learning	
	Advanced Topics	

Lecture contents

Recap: model-based and model-free RL

- Until now, knowledge of a model only in the simplest RL approaches.
 - Dynamic Programming for tabular RL, e.g., policy or value iteration.
 - We need to know $p(s'|s, a)$ to update our value estimate!
- All the advanced algorithms: **model-free RL**.
 - Collect experience by following the current policy π ,
 - then update our estimate of V , Q , A or π only from the data.

Advantages / disadvantages of model-free RL

- + No model knowledge required. All we need is access to the system / a black-box simulator.
- + No bias due to inaccurate model.

Today: Can we do better if we know the dynamics?

We often know the dynamics:

- Games (e.g., Chess, Go, Atari games).
- Easily modeled systems (e.g., car navigation).

Algorithm: Value iteration for estimating $\pi \approx \pi^*$.

while ($\Delta > \theta$):

$\Delta \leftarrow 0$

for $s \in \mathcal{S}$:

$V_{\text{old}}(s) \leftarrow V(s)$

$V(s) \leftarrow \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} p(s'|s, a) [r + \gamma V(s')]$

$\Delta \leftarrow \max(\Delta, |V_{\text{old}}(s) - V(s)|)$

+ Easier to implement.

— Data-hungry! Can be very sample-inefficient.

— Ignorant of prior knowledge.

Also, we can often **learn the dynamics**:

- System identification: fit unknown parameters of a known model.
- Learning: fit a general purpose model to observed transition data.

- Simulated environments (e.g., robotics, video games)

What is a model?

- Very generally, a **model** is an MDP tuple $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$, just as we have seen before. Most notably, we have
 - the dynamics model $p(s'|s, a)$,
 - the reward probability $r \sim p(\cdot|s, a)$.
- The state and action spaces $\mathcal{S}$ and $\mathcal{A}$ are usually assumed to be known.
- The discount factor is given or can be chosen by an expert.

Types of models

1. Probabilistic vs. deterministic

- probabilistic:

$$s' \sim p(\cdot|s, a), \quad r \sim p(\cdot|s, a).$$

- deterministic (a special case of the former):

$$s' = f(s, a), \quad r = \ell(s, a).$$

2. Known vs. unknown

- **True MDP:** p and r (or f and ℓ) are perfectly known a priori.
- **Approximated MDP:** p and r (or f and ℓ) need to be learned,

$$\tilde{p} \approx p, \tilde{r} \approx r \quad \text{or} \quad \tilde{f} \approx f, \tilde{\ell} \approx \ell.$$

Open-loop optimal control

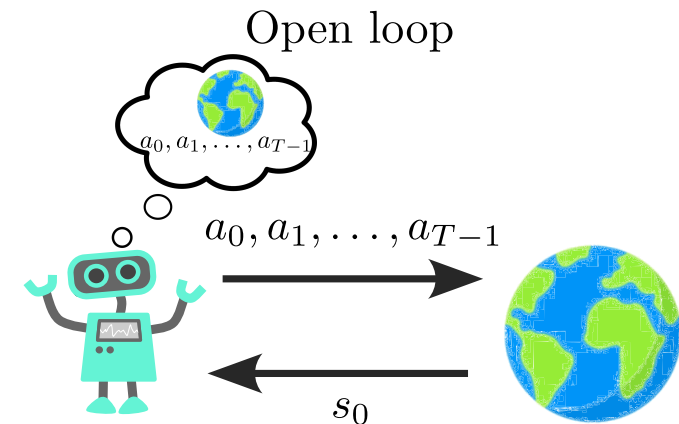
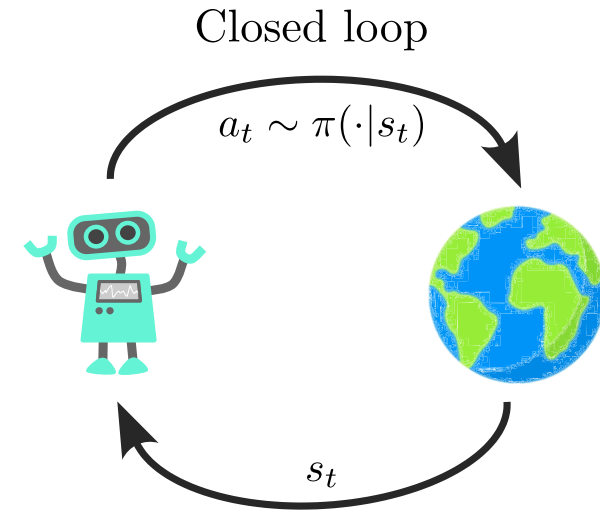
Planning & optimal control

- For now, we assume that the model
 - is known,
 - is fully deterministic.
- In the episodic case ($T < \infty$), the RL problem becomes

$$\min_{a_0, \dots, a_{T-1}} \sum_{t=0}^T \ell(s_t, a_t) = \min_{a_0, \dots, a_{T-1}} L(a) \quad (1)$$

subject to $s_{t+1} = f(s_t, a_t), t = 0, 1, \dots, T-1$.

- Eq. (1) is referred to as an **optimal control problem (OCP)**.
 - ℓ is the *stage cost*.
 - 💡 Closely related to the negative reward $-r$!
 - L is the *cost* that we seek to minimize.
 - 💡 Closely related to value V !
 - Optimal control is usually formulated as a *minimization problem*, e.g., the distance to a desired trajectory.
- Generally, optimal control refers to the **open-loop / planning** setting.



Only measure at s_0 , no feedback

Inspired by Sergey Levine's CS285 lecture.

The deterministic vs. the stochastic case

- In RL, we usually assume that we have a stochastic problem for various reasons:
 - stochastic dynamics,
 - stochastic actions/policies,
 - stochastic rewards.
- In the open-loop setting, stochastic optimal control is suboptimal!
⇒ Why?

Stochastic dynamics

$$p(s_0, \dots, s_T | a_0, \dots, a_{T-1}) = p(s_0) \prod_{t=0}^{T-1} p(s_{t+1} | s_t, a_t).$$

- The optimal action a_1 depends on s_1 , which we cannot know for certain at $t = 0$.
- The same is true for all following states and actions.

Some approaches for stochastic optimal control

- Certainty Equivalence (CE)
 - Simply replace all stochastic quantities by their expectations/mean values.
+ Computationally cheap.
— Completely ignores variance.
- Mean-variance / risk-sensitive optimization
 - define loss function composed of mean and variance:

$$\min_a \mathbb{E}[L] + \gamma \text{var}[L].$$

- + Accounts for variance.
- Need additional compute to estimate $\text{var}[L]$.

- Sample-path optimization
 - Monte-Carlo estimate of expected loss by multiple simulations with different noise realizations ω_i :

$$\min_a \frac{1}{N} \sum_{i=1}^N L(a, \omega_i).$$

+ Can estimate arbitrary distributions (beyond Gaussian).

— Compute grows with N .

Solving the OCP: single shooting

$$\min_{a_0, \dots, a_{T-1}} \underbrace{\sum_{t=0}^T \ell(s_t, a_t)}_{=L(a)} \quad \text{subject to} \quad s_{t+1} = f(s_t, a_t), \quad t = 0, 1, \dots, T-1. \quad (1)$$

A very common approach to solve the OCP (1) is via **single shooting**, where assume that the states s_t are implicitly defined by choosing the actions a_t .

💡 This holds for most systems of interest. The basis for this are existence-and-uniqueness theorems, e.g., by **Picard-Lindelöf**.

Single shooting

Initialize: Initial state s_0 , action sequence $a^{(0)} = (a_0^{(0)}, \dots, a_{T-1}^{(0)})$.

for $j = 0, 1, 2, \dots$:

 simulate the system following $a^{(j)}$ to get $s^{(j)} = (s_1^{(j)}, \dots, s_T^{(j)})$

 assess the performance by computing $L(a^{(j)})$

 gradient descent to update our actions:

$$a^{(j+1)} = a^{(j)} - \eta \nabla_a L(a^{(j)})$$

Remaining question:

How do we compute the gradient

$$\nabla_a L(a^{(j)})?$$

Gradients in single shooting (1)

$$\min_{a_0, \dots, a_{T-1}} \underbrace{\sum_{t=0}^T \ell(s_t, a_t)}_{=L(a)} \quad \text{subject to} \quad s_{t+1} = f(s_t, a_t), \quad t = 0, 1, \dots, T-1. \quad (1)$$

The procedure of computing the gradient is closely related to the backpropagation algorithm!

1. Turn the constraint into an objective via the Lagrangian formalism, where λ is the adjoint state:

$$\begin{aligned} \mathcal{L}(s, a, \lambda) &= \sum_{t=0}^T \ell(s_t, a_t) + \sum_{t=0}^{T-1} \lambda_{t+1}^\top (f(s_t, a_t) - s_{t+1}) \\ &= [\ell(s_0, a_0) + \lambda_1^\top f(s_0, a_0)] + [\ell(s_1, a_1) + \lambda_2^\top f(s_1, a_1) - \lambda_1^\top s_1] + \dots + \\ &\quad [\ell(s_t, a_t) + \lambda_{t+1}^\top f(s_t, a_t) - \lambda_t^\top s_t] + \dots + [\ell(s_T, a_T) + \lambda_T^\top s_T]. \end{aligned}$$

2. Necessary condition for optimality: Total derivative of $\mathcal{L}$ has to be zero:

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0, \quad \frac{\partial \mathcal{L}}{\partial s_t} = 0, \quad \frac{\partial \mathcal{L}}{\partial a_t} = 0.$$

Gradients in single shooting (2)

$$\mathcal{L}(s, a, \lambda) = \sum_{t=0}^T \ell(s_t, a_t) + \sum_{t=0}^{T-1} \lambda_{t+1}^\top (f(s_t, a_t) - s_{t+1})$$

3. State equation:

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = s_{t+1} - f(s_t, a_t) \stackrel{!}{=} 0 \quad \Leftrightarrow \quad s_{t+1} = f(s_t, a_t) \quad \text{for } t = 0, \dots, T-1.$$

4. Adjoint equation:

$$\frac{\partial \mathcal{L}}{\partial s_t} = \frac{\partial \ell}{\partial s_t} + \left(\frac{\partial f}{\partial s_t} \right)^\top \lambda_{t+1} - \lambda_t \stackrel{!}{=} 0 \quad \Leftrightarrow \quad \lambda_t = \frac{\partial \ell}{\partial s_t} + \left(\frac{\partial f}{\partial s_t} \right)^\top \lambda_{t+1} \quad \text{for } t = 0, \dots, T-1.$$

5. Final adjoint state ($t = T$): No influence on a future state, $\lambda_T = \frac{\partial \ell}{\partial s_T}$.

6. Action gradient:

$$\frac{\partial \mathcal{L}}{\partial a_t} = \frac{\partial \ell}{\partial a_t} + \left(\frac{\partial f}{\partial a_t} \right)^\top \lambda_{t+1} \quad \text{for } t = 0, \dots, T-1.$$

Gradients in single shooting (3)

Gradient calculation in the single shooting method for solving (1)

Given: Initial state s_0 , action sequence $a = (a_0, \dots, a_{T-1})$.

- **Forward simulation** of the state equation:

$$s_{t+1} = f(s_t, a_t), \quad t = 0, 1, \dots, T-1.$$

- “Initial condition” (= condition at final time T) for the adjoint equation: $\lambda_T = \frac{\partial \ell}{\partial s_T}$.
- **Backward simulation** of the adjoint equation:

$$\lambda_t = \frac{\partial \ell}{\partial s_t} + \left(\frac{\partial f}{\partial s_t} \right)^\top \lambda_{t+1}, \quad t = T-1, T-2, \dots, 1, 0.$$

- **Gradient calculation w.r.t. a :**

$$\frac{\partial \mathcal{L}}{\partial a_t} = \frac{\partial \ell}{\partial a_t} + \left(\frac{\partial f}{\partial a_t} \right)^\top \lambda_{t+1}, \quad t = 0, 1, \dots, T-1.$$

Notes

- The backward simulation is closely related to the backward pass in backpropagation.
- In modern packages such as PyTorch, JAX or TensorFlow, this procedure can be automated with autodiff, when appropriately implementing the forward pass.

Solving the OCP: full discretization

- Alternative to forward-backward: the full discretization
- Both s and a become optimization variables:

$$x = [s_0^\top, a_0^\top, s_1^\top, a_1^\top, \dots, a_{T-1}^\top, s_T^\top]^\top.$$

- Reformulation of the OCP (1)

$$\min_x F(x) = \sum_{t=0}^T \ell(s_t, a_t) \quad \text{s.t.} \quad C(x) = \begin{bmatrix} f(s_0, a_0) - s_1 \\ f(s_1, a_1) - s_2 \\ \vdots \\ f(s_{T-1}, a_{T-1}) - s_T \end{bmatrix} = 0 \quad (2)$$

- Total number of optimization variables when the state dimension is n and the action dimension is m : $(T + 1)n + Tm$.
- Solution via standard solvers for nonlinear, constrained optimization problems:
 - Require the gradients $\nabla_x F(x)$ and $\nabla_x C(x)$.
 - The latter is a very large matrix, but it is fortunately very sparse.

Comparison: Shooting vs. full discretization

	Shooting (1)	Full discretization (2)
Sensitivity to initialization	Very sensitive to poor initial guesses $a^{(0)}$.	Because we initialize the entire state trajectory s , the system doesn't have to be dynamically feasible during early solver iterations $\Rightarrow$ Significantly more stable for unstable dynamics.
Computational cost vs. convergence	Sequential: Fast iterations (one forward and backward pass), but may take many iterations to converge.	Full Discretization: Iterations computationally heavier, but standard Sequential Quadratic Programming (SQP) or Interior Point methods achieve quadratic convergence near the solution, requiring far fewer total iterations.
Constraint Handling	Limits on states like $s_{\min} \leq s_t \leq s_{\max}$ or actuations ($a_{\min} \leq a_t \leq a_{\max}$) can be handled easily. They are simply treated as bound constraints on elements of our vector x .	Handling state constraints requires challenging penalty functions or barrier methods.

The linear-quadratic case

The linear-quadratic case

Let's consider a special (yet very common) case

- **Linear dynamics:**

$$s_{t+1} = As_t + Ba_t, \quad \text{with } A \in \mathbb{R}^{n \times n} \text{ and } B \in \mathbb{R}^{n \times m}.$$

- **Quadratic losses:**

$$L(s, a) = \left(\sum_{t=0}^{T-1} s_t^\top Q s_t + a_t^\top R a_t \right) + s_T^\top Q_f s_T,$$

with matrices $Q, Q_f \in \mathbb{R}^{n \times n}$ penalizing the (terminal) state and $R \in \mathbb{R}^{m \times m}$ penalizing the control cost.

Additional conditions

- The matrix Q is *positive semidefinite*:

$$s^\top Q s \geq 0 \quad \forall s \in \mathbb{R}^n.$$

- The matrix R is *positive definite*:

$$a^\top R a \geq 0 \quad \forall a \in \mathbb{R}^m.$$

- These ensure existence and uniqueness of a minimizer s^*, a^* .

Linear-quadratic optimal control problem

$$\begin{aligned} & \min_{s,a} \left(\sum_{t=0}^{T-1} s_t^\top Q s_t + a_t^\top R a_t \right) + s_T^\top Q_f s_T \\ \text{subject to } & s_{t+1} = A s_t + B a_t, \quad t = 0, 1, \dots, T-1. \end{aligned} \tag{3}$$

Removing the constraints

Let's explicitly calculate $s_1, s_2, \dots$:

$$s_1 = As_0 + Ba_0$$

$$s_2 = As_1 + Ba_1 = A(As_0 + Ba_0) + Ba_1 = A^2s_0 + ABa_0 + Ba_1$$

$$s_3 = As_2 + Ba_2 = A(A^2s_0 + ABa_0 + Ba_1) + Ba_2 = A^3s_0 + A^2Ba_0 + ABa_1 + Ba_2$$

General rule:
$$s_t = A^t s_0 + \sum_{i=0}^t A^{t-i-1} B a_i$$

Explicitly:

$$\underbrace{\begin{bmatrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ \vdots \\ s_T \end{bmatrix}}_{\hat{s} \in \mathbb{R}^{(T+1)n}} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & B \\ 0 & 0 & 0 & \dots & 0 \\ AB & B & 0 & \dots & 0 \\ A^2B & AB & B & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A^{T-1}B & A^{T-2}B & A^{T-3}B & \dots & B \end{bmatrix}}_{G \in \mathbb{R}^{(T+1)n \times Tm}} \underbrace{\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_{T-1} \end{bmatrix}}_{\hat{a} \in \mathbb{R}^{Tm}} + \underbrace{\begin{bmatrix} I \\ A \\ A^2 \\ A^3 \\ \vdots \\ A^T \end{bmatrix}}_{H \in \mathbb{R}^{(T+1)n}} s_0$$

In short:
$$\hat{s} = G\hat{a} + Hs_0$$

Reformulating the cost function

- Next, we stack our penalty matrices such that they match the dimensions of S and U :

$$\hat{Q} = \begin{bmatrix} Q & & \\ & \ddots & \\ & & Q \\ & & & Q_f \end{bmatrix} \in \mathbb{R}^{(T+1)n \times (T+1)n}, \quad \hat{R} = \begin{bmatrix} R & & \\ & \ddots & \\ & & R \end{bmatrix} \in \mathbb{R}^{Tm \times Tm}.$$

- We can now insert this into the linear-quadratic OCP (3):

$$\min_{s,a} \left(\sum_{t=0}^{T-1} s_t^\top Q s_t + a_t^\top R a_t \right) + s_T^\top Q_f s_T \quad \text{s.t.} \quad s_{t+1} = A s_t + B a_t, \quad t = 0, 1, \dots, T-1$$

becomes

$$\min_{\hat{s}, \hat{a}} \hat{s}^\top \hat{Q} \hat{s} + \hat{a}^\top \hat{R} \hat{a} \quad \text{s.t.} \quad \hat{s} = G \hat{a} + H s_0.$$

- **Question:** Do we have to consider $\hat{s}$ and $\hat{a}$ independently?
⇒ Let's simply insert the constraints!

$$\min_{\hat{a}} (G\hat{a} + Hs_0)^\top \hat{Q} (G\hat{a} + Hs_0) + \hat{a}^\top \hat{R} \hat{a} \quad (4)$$

Solving the linear OCP

$$\begin{aligned}
 & \min_{\hat{a}} (G\hat{a} + Hs_0)^\top \hat{Q} (G\hat{a} + Hs_0) + \hat{a}^\top \hat{R} \hat{a} \\
 &= \min_{\hat{a}} (\hat{a}^\top G^\top + s_0^\top H^\top) \hat{Q} (G\hat{a} + Hs_0) + \hat{a}^\top \hat{R} \hat{a} \\
 &= \min_{\hat{a}} \hat{a}^\top G^\top \hat{Q} G \hat{a} + 2\hat{a}^\top G^\top \hat{Q} H s_0 + s_0^\top H^\top \hat{Q} H s_0 + \hat{a}^\top \hat{R} \hat{a} \\
 &= \min_{\hat{a}} \hat{a}^\top (G^\top \hat{Q} G + \hat{R}) \hat{a} + 2\hat{a}^\top G^\top \hat{Q} H s_0 + s_0^\top H^\top \hat{Q} H s_0
 \end{aligned}$$

Some linear algebra

$$\begin{aligned}
 (G\hat{a})^\top &= \hat{a}^\top G^\top \\
 s_0^\top H^\top \hat{Q} G \hat{a} &= \hat{a}^\top G^\top \hat{Q} H s_0
 \end{aligned}$$

- This is a quadratic polynomial in $\hat{a}$!
- Since Q and R (and thus, $\hat{Q}$ and $\hat{R}$) are positive (semi-)definite, the loss function is strictly convex $\Rightarrow$ unique minimizer.
- Take the derivative with respect to $\hat{a}$:

$$\frac{d}{d\hat{a}} \left(\hat{a}^\top (G^\top \hat{Q} G + \hat{R}) \hat{a} + 2\hat{a}^\top G^\top \hat{Q} H s_0 + s_0^\top H^\top \hat{Q} H s_0 \right) = 2(G^\top \hat{Q} G + \hat{R}) \hat{a} + 2G^\top \hat{Q} H s_0.$$

- Setting the gradient to zero yields a linear system (depending on the initial condition s_0):

$$\boxed{\left(G^{\top} \hat{Q} G + \hat{R}\right) \hat{a} = -G^{\top} \hat{Q} H s_0.} \quad (5)$$

Example: Control of a forklift (1)

- Consider a simple vehicle driving on a plane (with mass m and friction d):

$$s = \begin{bmatrix} p_1 \\ v_1 \\ p_2 \\ v_2 \end{bmatrix}, \quad \dot{s} = \frac{ds}{dt}(t) = A_c s(t) + B_c a(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -d/m & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -d/m \end{bmatrix} s(t) + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} a(t).$$



- This can be translated into a discrete-time system with time step Δt :

$$A = \exp(A_c \Delta t), \quad B = \left(\int_0^{\Delta t} \exp(A(\Delta t - \tau)) d\tau \right) B_c \quad \Rightarrow \quad s_{t+1} = A s_t + B a_t.$$

- Control objective: Steer the forklift from $s_0 = [2 \ 0 \ 0 \ 0]^T$ to the origin with a small penalty on the control:

$$\min_{s,a} \sum_{t=0}^T \|s_t\|_2^2 + \alpha \|a_t\|_2^2 \quad \text{s.t.} \quad s_{t+1} = A s_t + B a_t.$$

Example: Control of a forklift (2)

- Define individual control matrices:

$$Q = \begin{bmatrix} 1 & & & \\ & 0 & & \\ & & 1 & \\ & & & 0 \end{bmatrix}, \quad Q_f = 10 \cdot Q, \quad R = \begin{bmatrix} 0.01 & \\ & 0.01 \end{bmatrix}.$$

- Assemble the matrices $\hat{Q}$ and $\hat{R}$. With $T = 50$, we get

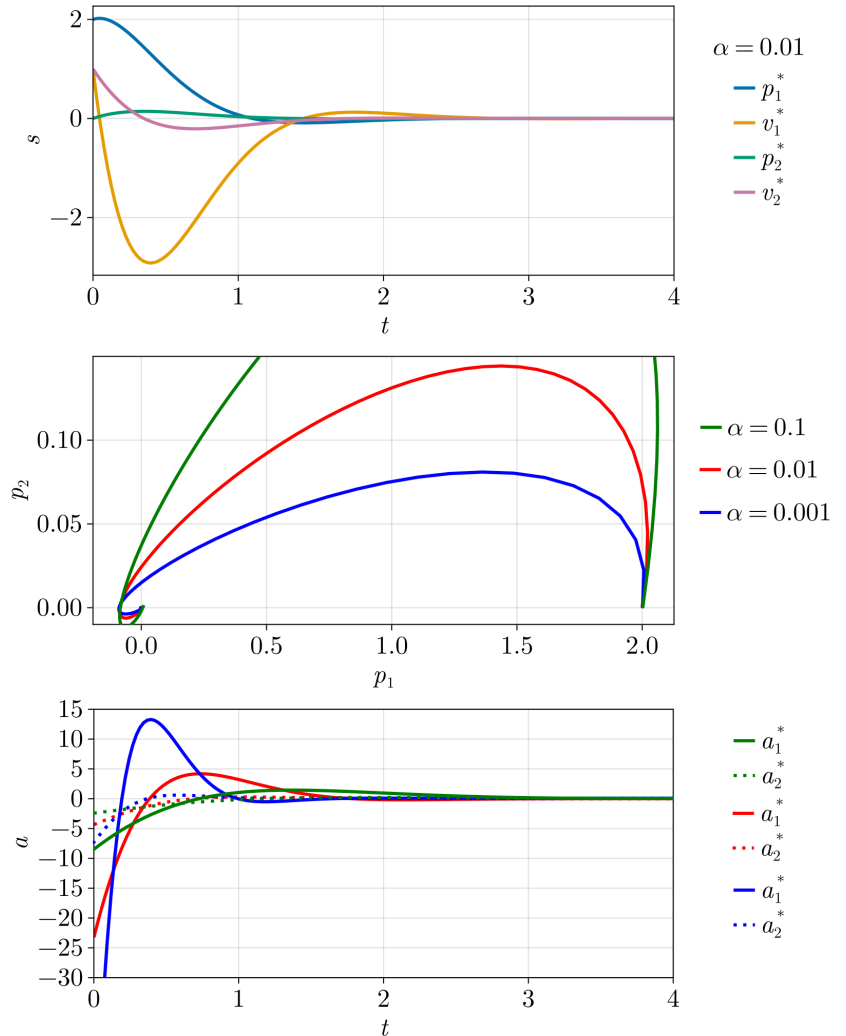
$$\hat{Q} \in \mathbb{R}^{204 \times 204} \quad \text{and} \quad \hat{R} \in \mathbb{R}^{100 \times 100}.$$

- Given the initial condition s_0 , solve the linear system

$$(G^\top \hat{Q} G + \hat{R}) \hat{a} = -G^\top \hat{Q} H s_0. \quad (5)$$

- Simulate dynamics to get optimal trajectory s^* :

$$s_{t+1}^* = A s_t^* + B a_t^* \quad \text{or} \quad \hat{s}^* = G \hat{a}^* + H s_0.$$



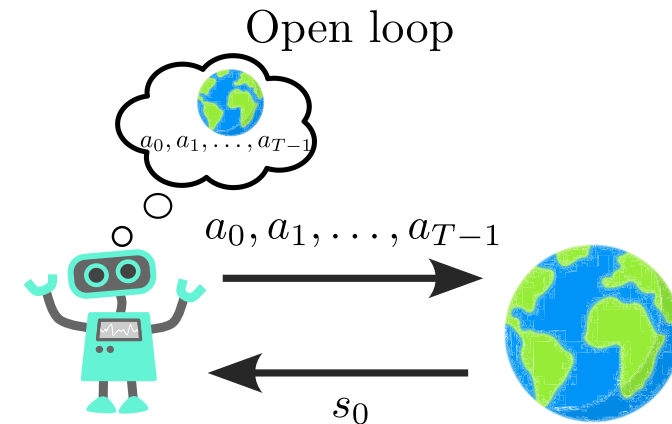
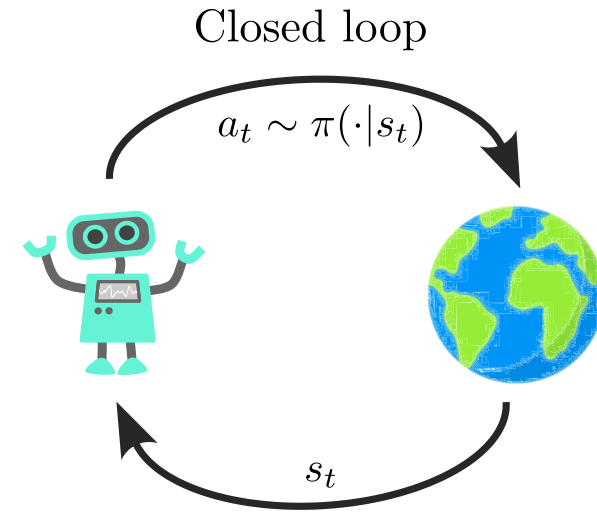
Closed-loop control

Model predictive control (MPC)

- Given a model, solving open-loop OCPs is a well-established field.
- But there are also **drawbacks** in the application to real systems.
 - Unforeseen events cannot be accounted for.
 - External disturbances/uncertainties are hard/impossible to include.
 - Smalles inaccuracies scale exponentially over time.
- **Question:** Can we close the loop around OCPs?

Model predictive control (MPC)

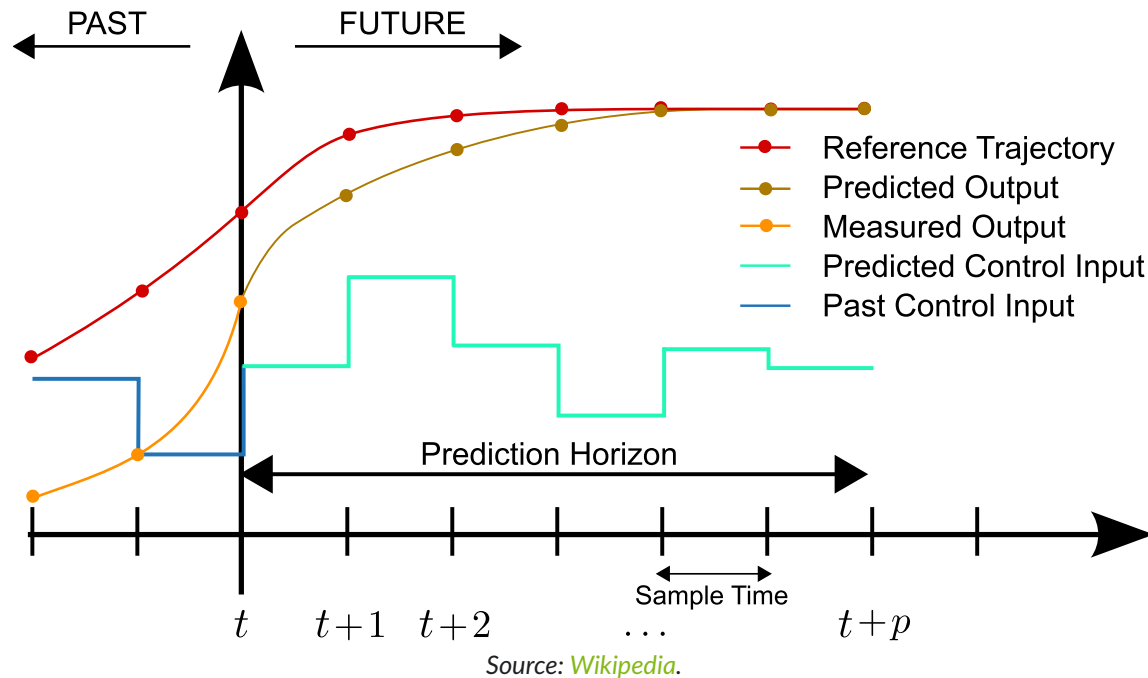
- Let us solve the open-loop problem repeatedly on a shorter horizon $p \leq T$.
- The “feedback” is setting the initial condition of the OCP.



Only measure at s_0 , no feedback

Inspired by Sergey Levine's CS285 lecture.

MPC scheme



The MPC loop

1. Measure system state s_t and use it as initial state for the OCP (1).
2. Solve (1) using a model of the real system over a prediction horizon of length p .
3. Apply first entry of a^* (i.e., a_0^*) to the real system.
4. Advance real system by one time step.
5. Repeat.

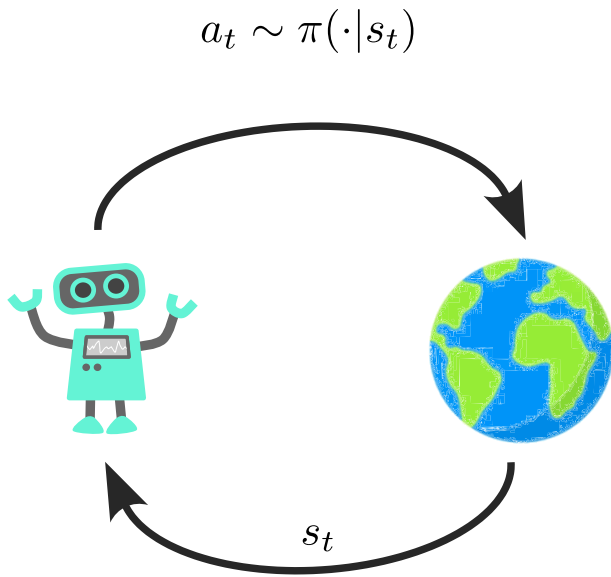
Pros and cons

- + Very robust.
- + No offline learning phase (if model known).
- + Easy to include state constraints.
- + Lots of theory. (e.g., (Grüne and Pannek 2017))

- Real-time capability (in particular for nonlinear models).
- Control performance depends on model quality.

Relation to reinforcement learning

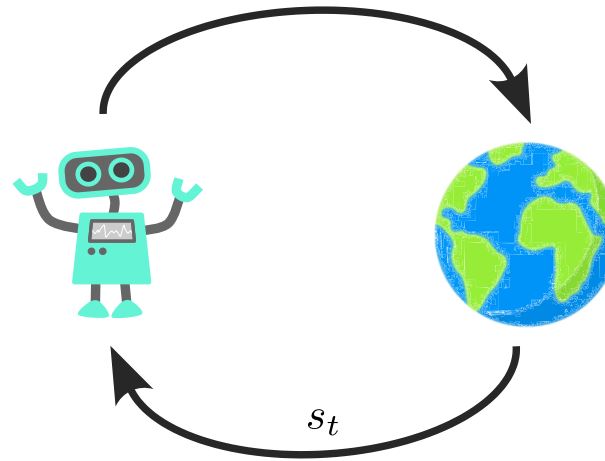
Reinforcement Learning



VS.

Model Predictive Control

$$a_t = \arg \min_{\hat{a}} \sum_{\tau=0}^p \ell(\hat{s}_{\tau}, \hat{a}_{\tau})$$
$$\text{s.t. } \hat{s}_{\tau+1} = f(\hat{s}_{\tau}, \hat{a}_{\tau})$$
$$\hat{s}_0 = s_t$$



Trade-offs

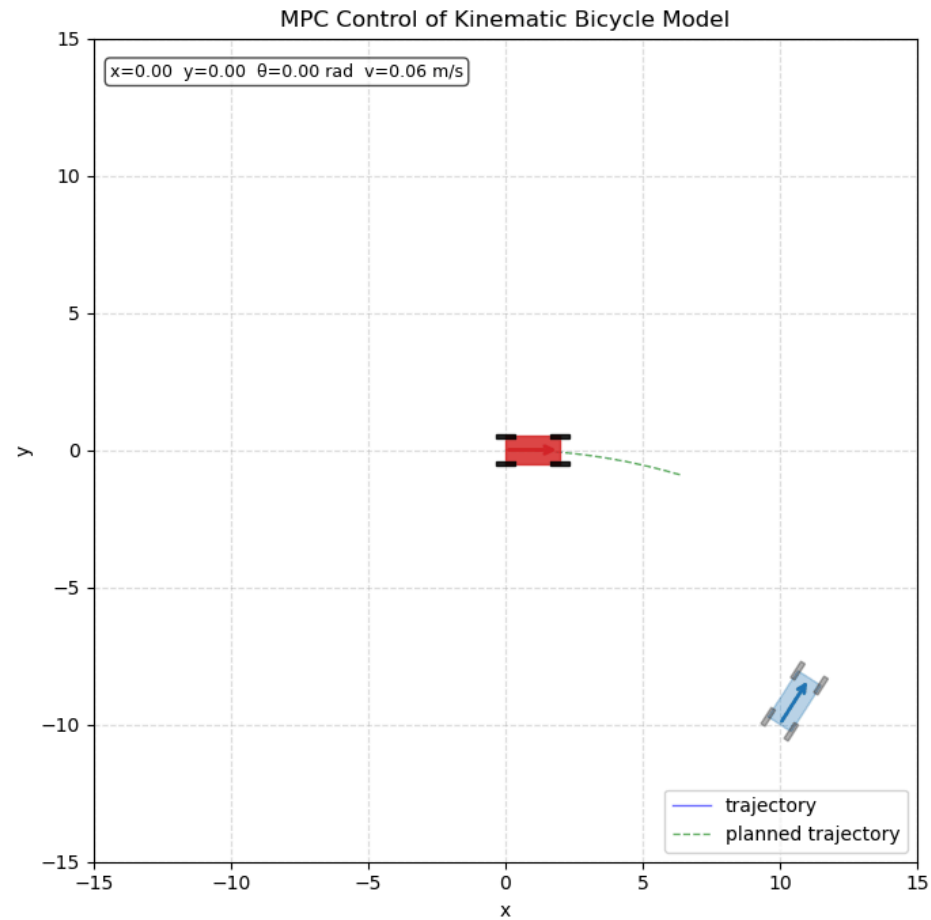
- Learning a policy offline vs. solving an OCP online.
- Expensive training vs. expensive inference.

Approaches in between

- Explicit MPC
- Differentiable predictive control (DPC)
- Monte-Carlo tree search (MCTS)

Example: Autonomous driving

Trajectory planning plus MPC for following that trajectory



Explicit MPC

Differentiable predictive control (DPC)

References

Grüne, L., and J. Pannek. 2017. *Nonlinear Model Predictive Control*. 2nd ed. Springer International Publishing.